

HARSHVARDHAN TAKAWALE

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RESEARCH INTERESTS

Spatial Acoustics, Physics-informed Machine Learning, Multimodal imaging, RF sensing, Ambient Computing, Ubiquitous Healthcare, Low-power Learning Systems

EDUCATION

University of Maryland, College Park

2022 -

Ph.D. in Computer Science

Advisor - Dr. Nirupam Roy

Birla Institute of Technology and Science, Pilani

2016 - 2020

Bachelor of Engineering in Electrical and Electronics

Minor in Data Science

PUBLICATIONS

- **INFER : Learning Implicit Neural Frequency Response Fields for Confined Car Cabin** ([link](#))
Harshvardhan Takawale, Nirupam Roy, Phil Brown
(ICML 2026)
- **Spectral Domain Neural Reconstruction for Passband FMCW Radars** ([link](#))
Harshvardhan Takawale, Nirupam Roy
(Under Review - NeurIPS 2026)
- **LITE: Loss-resilient Immersive Telepresence with Multi-modal Semantics**
Ruizhi Cheng, Harshvardhan Takawale, Nan Wu, Nirupam Roy, Sennur Ulukus, Matteo Varvello, Eugene Chai, Bo Han
(SIGCOMM 2026)
- **Sparse is Fast: Accelerating Synthetic Aperture Sonar via Reduced Sampling** ([link](#))
Yang Bai, Harshvardhan Takawale, Aratrik Ghosh, Nirupam Roy
(GetMobile 2026)
- **Towards Detecting Auditory Attention from in-Ear Muscle Contractions using Commodity Earbuds** ([link](#))
Harshvardhan Takawale, Yang Liu, Khaldoun Al-Naimi, Fahim Kawsar, Alessandro Montanari
(ICASSP 2025)
- **Learning Speaker-Listener Mutual Head Orientation by Leveraging HRTF and Voice Directivity on Headphones** ([link](#))
Harshvardhan Takawale, Nirupam Roy
(ICASSP 2024)

- **Scribe: Simultaneous Voice and Handwriting Interface** ([link](#))
Yang Bai, Irtaza Shahid **Harshvardhan Takawale** and Nirupam Roy
(IMWUT 2024)
- **Structure Assisted Spectrum Sensing for Low-power Acoustic Event Detection** ([link](#))
Nakul Garg, **Harshvardhan Takawale**, Yang Bai, Irtaza Shahid and Nirupam Roy
IASA CPS IoT Week 2023
- **Poster: Natural Voice Interface for the Next Generation of Smart Spaces** [link](#)
Yang Bai, Nakul Garg, **Harshvardhan Takawale**, Anupam Das and Nirupam Roy
24rd HotMobile 2023
- **A Seamless Second-Factor Verification as a Mobility-Service for Future Commute Stations** ([link](#))
Harshvardhan Takawale, Jay Prakash and Tony Quek
23rd HotMobile 2022
- **Malware Detection on Highly Imbalanced Data through Sequence Modeling** ([link](#))
Rajvardhan Oak, Min Du, David Yan, **Harshvardhan Takawale** and Idan Amit
12th ACM Workshop on Artificial Intelligence and Security, Nov. 2019
- **Fault-Tolerant Routing Algorithm for Mesh based NoC using Reinforcement Learning** ([link](#))
Jagadheesh Samala, **Harshvardhan Takawale**, Yash Chokhani, P Veda Bhanu, J Soumya
24th International Symposium on VLSI Design and Test (VDATE), July. 2020
- **Talos App: On-Device Machine Learning Using TensorFlow to Detect Android Malware** ([link](#))
Harshvardhan Takawale and Abhishek Thakur
5th International Conference on Internet of Things: Systems, Management & Security, Oct. 2018

PAST RESEARCH EXPERIENCE

Dolby Laboratories, San Francisco

Jun 2026 - Aug 2026

Ph.D. Research Intern | Supervisor : **Avery Bruni**

- Sparse Sound-Field Reconstruction via a Learned Generative Field Prior with Diffusion Posterior Sampling and Active Microphone Placement

Dolby Laboratories, San Francisco

Jun 2025 - Aug 2025

Ph.D. Research Intern | Supervisor : **Dr. Phil Brown**

- Modelling Car Cabin Acoustic Frequency Response Fields with Implicit Neural Representation

Nokia Bell Labs, Cambridge, UK

Jun 2024 - Aug 2024

Ph.D. Research Intern | Supervisor : **Dr. Fahim Kawsar**, Dr. Alessandro Montanari

- Detecting Auditory Attention from in-Ear Muscle Contractions using Commodity Earbuds

Silence Laboratories @ SUTD, Singapore

Jul 2020 - Jul 2022

Lead Researcher | Founders : **Prof. Tony Q. S. Quek**, Dr. Jay Prakash, Andrei Bytes

- Co-location based Seamless Authentication for Mobility Services
- Landmark extraction from infrastructure for indoor mobility claim verification
- Real Time Gesture Tracking for Passwordless Login

Cyber Security Research Centre @ NTU, Singapore

Jan 2020 - Jul 2020

Research Intern | Supervisor : **Prof. Thambipillai Srikanthan**, Asst. Prof. Lam Siew Kei

- Side-Channel Assisted Anomaly Detection for Embedded Platforms (Undergraduate Thesis)

Electrical Engineering & Computer Sciences @ U.C. Berkeley

Jun 2019 - Sep 2019

Remote Researcher | Lab PI : **Prof. Dawn Song** | Project head : Dr. Min Du

- Malware Detection on Highly Imbalanced Data through Sequence Modeling

School of Computer Science @ University of Windsor, Canada

May 2019 - Aug 2019

Security Research Intern | Supervisor : **Assoc. Prof. Saeed Samet**

- Integrating Continuous Authentication into the Personal Health Record Applications

PATENTS

- **Spatial Acoustic Perception on Miniature Computing Devices.**

Yang Bai, Nakul Garg, Nirupam Roy, Ayushi Mishra, Priyadarshan Narayanasamy,

Harshvardhan Takawale

US Patent

- **Auditory Attention Detection From In-Ear Muscle Contractions Using Commodity Earbuds.**

Harshvardhan Takawale, Yang Bai, and Alessandro Montanari

US Patent (Technology Disclosure Submitted)

- **Second Factor Authentication for Mobility: Ways to identify friendly frauds**

Jay Prakash, **Harshvardhan Takawale** and Tony Quek

Singapore Patent (Technology Disclosure Submitted)

AWARDS AND ACHIEVEMENTS

- Selected for **Financial Aid Award** at ICML 2026.
- Selected for **Dolby University Graduate Researcher Funding**.
- Selected for **Gold Reviewer Award** at ICML 2026.
- Selected for **NSF I-Corps** program as **Entrepreneurial Lead**.
- Selected as a **Principal Investigator** for **Seed Award Program** by Artificial Intelligence Interdisciplinary Institute at Maryland (AIM).
- University of Maryland Graduate School **Dean's Fellowship** for the academic year 2022-23 and 2023-24
- Awarded a fully-funded semester-long internship at Nanyang Technological University to pursue undergraduate thesis
- **Mitacs GRI Scholar** - Selected amongst 700 students in a pool of 200,000 applicants for a fully funded research internship in Canada for the Summer of 2019

- Awarded with **National Talent Search Examination Scholarship** by Indian govt. that is given to 1000 students out of the total 5,000,000+ participants in the entire country.
- Placed first in Urban Livability Challenge 2019 Hackathon by eGovernments Foundation.
- Runner-up at the Megathon-2018 Hackathon
- National Winner of CBSE Heritage India Quiz by Indian govt., with 8000+ participating teams

TEACHING

- CMSC436 Programming Handheld Systems, Spring'25
Instructor: Prof. Michael Marsh
- CMSC422 Introduction to Machine Learning, Spring'23
Instructor: Prof. Mohammad Nayeem Teli
- CMSC122 Introduction to Computer Programming via the Web, Fall'22
Instructor: Prof. Pedram Sadeghian

SERVICES

- Web Chair, HumanSys 2025
- Publicity and Social Media Chair, HumanSys 2024
- Program Committee, ACM S3 2023
- Artifact Evaluation committee, ACM MobiSys 2023
- Replicability committee, ACM WiSec 2022